MODEL QUESTION PAPER
DIPLOMA IN INFORMATION TECHNOLOGY
S3.04-Software Engineering

Time : 3 Hour *Max.Marks : 75*

PART A

I. Answer **all** questions in one word or one sentence. (9 x 1 = 9 Marks)

1	Name the system where the software controls some hardware device and is embedded in that device.	M1.01	R
2	Name the first model of software development process derived from engineering process models used in large military systems.	M1.03	R
3	Write the three types of software components that are frequently reused.	M1.05	R
4	Name the alias of Rapid software development.	M 2.01	R
5	Name the agile method that provides framework for organizing agile projects.	M 2.07	R
6	Write three instances of UML diagram types	M 3.04	R
7	Write the name of the approach of software development, in which a system is represented as a set of models that can be automatically transformed to executable code	M 3.07	R
8	Define the testing method in which we test individual objects and methods.	M 4.06	U
9	Write the three different types of user testing.	M 4.10	R

PART B

II. Answer any **eight** questions from the following. Each question carries 3 marks.

(8 x 3 = 24 Marks)

1	Describe the areas of software engineering where standards are bound by professional responsibility rather than laws.	M1.01	U
2	Explain why the waterfall model is applicable for only some types of systems.	M1.03	U
3	Explain how agile approach is different from plan driven approach.	M2.03	U
4	Discuss the advantages and disadvantages of pair programming.	M2.06	R
5	Explain the need of refactoring in agile software development.	M2.01	U
6	Compare the functional and non-functional requirements of software system requirements.	M3.01	U
7	Describe the different types of checks carried out on the requirements validation process.	M3.03	U

8	Explain the four fundamental configuration management activities..	M4.05	R
9	Describe the test-driven development approach to program development. State the various steps involved in the process.	M4.08	U
10	State and explain the various levels of testing process that decide whether the software is good enough to be deployed..	M4.10	R

PART C

Answer **all** questions. Each question carries 7 marks. (6 x 7 = 42 Marks)

III	Explain three types of software components that are frequently used in re-use-orientedapproaches of software engineering	M 1.05	U
	OR		
IV	Explain the various software process methods	M1.03	R
V	Justify the statement “It is difficult to introduce agile methods into large companies”	M2.09	U
	OR		
VI	Explain how the agile methodology will be useful for students to complete a major project	M2.02	U
VII	Explain the three types of abstract system model recommended by Model driven architecture.	M3.08	R
	OR		
VIII	Describe the necessary guidelines to minimize misunderstandings while writing software requirements in natural language	M3.02	U
IX	Using the UML graphical notation for object classes, design the following object classes, identifying attributes and operations. (a) A messaging system on a mobile(cell)phone or tablet (b) A printer for a personal computer (c) A personal music system (d) A bank account (e) A library catalogue	M4.03	U
	OR		
X	Explain the idea of integrating components for component interface testing.	M4.07	U

XI	Explain the two levels of abstraction at which software architectures can be designed	M3.10	R
	OR		
XII	Illustrate the use-case modelling with UML diagram support using a suitable example	M3.05	U
XIII	The goal of verification and validation processes is to establish confidence that the software system is fit for purpose. State the various factors on which the level of required confidence depends	M4.06	U
	OR		
XIV	Explain the different types of testing processes and validation in detail	M4.10	R

Mark Distribution

Module	hr /Module	Marks / Module (hi/ΣHi) * 123 (±5%)	Type of Questions							
			Part A		Part B		Part C		Total	
			No. of Questions	Marks	No. of Questions	Marks	No. of Questions	Marks	No. of Questions	Marks
1	11	31.465	3	3	2	6	2	14	7	25
2	10	28.604	2	2	3	9	2	14	7	36
3	12	34.326	2	2	2	6	4	28	8	36
4	10	28.605	2	2	3	9	4	28	9	26
Total	**43**	**123**	9	9	10	30	12	84	31	123

Cognitive Level Mark Distribution

Cognitive Level	Marks	% of Marks
Remembering	45	37
Understanding	78	63
Applying		
Analysing		
Evaluating		
Creating		
Total	123	100